Fiscal Year:	FY 2005	**Task Last Updated:**	FY 01/28/2005
PI Name:	Wichman, Indrek Sven Ph.D.		
Project Title:	Investigation of Diffusion Flame Tip Instability in Microgravity		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Surface combustion, shs, fire safety and smoldering		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Web Page:			
City:	East Lansing	**State:**	MI
Zip Code:	48824-1226	**Congressional District:**	8
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	97-HEDS-01
Start Date:	04/01/2004	**End Date:**	01/04/2008
No. of Post Docs:	0	**No. of PhD Degrees:**	2
No. of PhD Candidates:	1	**No. of Master' Degrees:**	3
No. of Master's Candidates:	3	**No. of Bachelor's Degrees:**	4
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA GRC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	Note: renewed 4 yrs		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Olson, Sandra L. (NASA Glenn Research Center)		
Grant/Contract No.:	NNC04AA50A		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The goal of ATHINA is to examine the transition of flame fronts to flamelet spread. This occurs in low gravity near-limit conditions. In addition, for Fire Safety reasons it is necessary to examine the reverse transition, from flamelet spread to active flame spread. This might occur when a fresh quantity of air enters the region, and poses a serious fire hazard aboard spacecraft. ATHINA thus has a basic science component and a definite Fire Safety component. The objective of ATHINA is to conduct basic science and applied engineering research on the flame-to-flamelet (and vice-versa) transition process. Experimental, theoretical, and numerical means are to be employed to that end. Fire Safety in space has become a larger concern as NAS has become more mission-oriented. The significance of ATHINA appears in both its basic science implications (the flamelet regime being a troublesome intermediate between flaming and quenching in which the flame very capably "adapts" to maximize its survival in adverse conditions), and in its engineering implications (being directly concerned with fire and flame spread in near-limit microgravity environments).		

	The importance to Fire Safety is the largest element of significance in this project.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The device developed for ATHINA (earth-based), alternatingly called a "hele-Shaw" facility and a "Turing" facility (thus presently unnamed) is useful for assessing the fire hazard in tight spaces, such as narrow ducts and walls, where fires may be initiated.
Task Progress:	Progress has occurred in four areas: 1) A new "Hele-Shaw" or "Turing" facility has been built and tested at NASA. It is capable of examining high oxygen concentration environments. Extensive tests have been performed at various gap spacings and flow velocities. Two different oxygen levels have been used. 2) Recent analysis of the test data has used population demographics techniques to examine flamelet bifurcations and extinctions, as well as surface coverage by flamelets, flamelet average size, and flamelet average spacing. 3) The drops in the 5 sec "zero-g" facility inceased to about 200 in the period of this report. Results obtained are currently being examined both statistically and mechanistically. 4) Modeling work at MSU continues with emphasis on two areas: (i) The 3-D simulation of processes occurring in the "Hele'Shaw" configuration; (ii) The 2-D rigorous examination of flame ignition and spread over thin combustible solid materials. In addition, theoretical work is continuing on the methods of analysis of the data on tests. Many additional techniques have been developed in the past year and are currently being implemented. These involve methods to estimate oxygen consumption rate and flamelt heat release rate.
Bibliography Type:	Description: (Last Updated: 10/16/2017)
Dissertations and Theses	Tanaya S. "Examination of a Simulated Micro-Gravity Device for Evaluating Flame Instability Transitions and Flame Spread over Thin Cellulosic Fuels." M.S. Thesis, Michigan State University, August 2004. , Aug-2004
Presentation	Long, Y., Wichman, I. S. "3-D Numerical Simulation of Flames Spreading over Thin Cellulosic Paper" Proceedings of the Central States Section Meeting of the Combustion Institute Mar-2004
Presentation	Long, Y., Wichman, I. S. "Ignition and Transition Process of Flames Spreading over Thin Cellulosic Fuel" Proceedings of the Central States Section meeting of the Combustion Institute Mar-2004